

A Novel Portable Iris Recognition System and Usability Evaluation

Youngkyoon Jang, Byung Jun Kang, and Kang Ryoung Park*

Abstract: We propose a new portable iris recognition system. Because existing portable iris systems use customized embedded processing units, they are limited in ability to expand to other applications, and they have low processing power. To overcome such problems, we propose a new portable iris recognition system consisting of a conventional ultra-mobile personal computer (UMPC), a small universal serial bus (USB) iris camera, and near-infrared (NIR) light illuminators. In general, portable iris systems produce considerable optical blurring. Although auto-focusing motor-driven lenses can be used to overcome it, they are too bulky to be used in a small-sized portable iris system. Therefore, we adopt an iris image restoration algorithm which performs at real-time speed. And by using a conventional UMPC as a processing unit, our portable iris system is more extensible than previous systems. In general, the performance of iris recognition has been mainly evaluated based on the quantitative metrics such as EER (Equal Error Rate), ROC (Receiver Operational Characteristics) curve or recognition time. We propose a new performance measuring method based on qualitative metrics. That is usability evaluation including user acceptance, convenience, satisfaction and resistance.

Keywords: Biometrics, iris recognition, portable iris system, usability evaluation.

1. INTRODUCTION

Recently, biometrics has gained popularity in many kinds of physical and logical security applications. It uses unique human features including the face, fingerprints, the iris, palm veins, speech and gait recognition [1-4]. Iris recognition identifies individuals based on unique iris texture patterns between the black pupil and white sclera. A conventional procedure for an iris recognition system is shown in Fig. 1. First, the system captures an iris image with a camera and a near infra-red (NIR) light illuminator. Preprocessing steps are then carried out, including iris segmentation and image normalization. Iris segmentation means that iris regions are detected from captured iris images, and occlusions such as eyelids, eyelashes, and specular reflection (SR) are removed. In the feature extraction step, iris feature codes are generated based on a wavelet filter, such as a

Gabor or Daubechies filter, in the detected iris region. The iris codes are then enrolled or matched with previously enrolled codes.

Due to its accuracy and reliability, many iris recognition systems have been commercialized such as IrisAccess 3000 and 4000 [5], and IRISPASS-M [6] and BM-ET200 [7]. They have mainly been used for door access control in an indoor environment, in a fixed position. To increase the mobility of iris recognition systems, portable iris recognition systems have been commercialized such as the IRISPASS-H [8], Pier™ 2.4 [9], and HIIDE™ Series 4 [10]. However, they use customized embedded systems which have limited extensibility to other applications [11], and have low processing power. For example, the Pier™ 2.4 system has 133 a MHz central processing unit (CPU), and the HIIDE system has a 533 MHz CPU [9,10].

To overcome these problems, we propose a new portable iris recognition system consisting of a conventional ultra-mobile personal computer (UMPC, model VGN-UX17LP of Sony Corp.) [12], a small universal serial bus (USB) iris camera, and near-IR (NIR) illuminators. By using the conventional UMPC as a processing unit, our system can be more extensible.

In Section 2, we describe the factors affecting the performance of the iris system. In Section 3, we show the proposed portable iris system. Experimental results are explained in Section 4. Conclusions are described in Section 5.

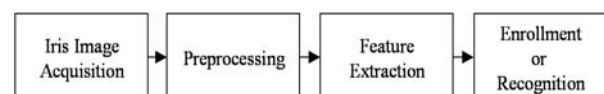


Fig. 1. Overall procedure of conventional iris recognition systems.

Manuscript received May 21, 2008; revised May 20, 2009; accepted August 15, 2009. Recommended by Editorial Board member Sooyong Lee under the direction of Editor Young-Hoon Joo. This work was supported by the Korea Science and Engineering Foundation (KOSEF) through the Biometrics Engineering Research Center (BERC) at Yonsei University [R112002105070020 (2009)].

Youngkyoon Jang is with the Department of Information and Communication, Gwangju Institute of Science and Technology (GIST), 1, Oryong-dong, Buk-gu, Gwangju, Korea (e-mail: yjang@gist.ac.kr).

Byung Jun Kang is with Electronics and Telecommunications Research Institute (ETRI), 138 Gajeongno, Yuseong-gu, Daejeon, Korea (e-mail: kangbyj@etri.re.kr).

Kang Ryoung Park is with the Dept. of Electronics Engineering, Dongguk University, Pildong-3ga, Jung-gu, Seoul, Korea (e-mail: parkgr@dongguk.edu).

* Corresponding author.

2. FACTORS AFFECTING IRIS SYSTEM PERFORMANCE

There have been many performance tests on biometric systems [13,14]. Tests sponsored by the U.S. Department of Homeland Security (DHS) showed the recognition performance of commercial biometrics systems including face, fingerprint, hand vein, voice, and iris recognition [13]. It showed that commercial iris recognition systems could provide a false acceptance rate (FAR) of about 10^{-6} % and a false rejection rate (FRR) of about 1% [13,15]. Other tests sponsored by the British National Physics Lab (NPL) showed that the recognition accuracy of commercial iris recognition systems was better than other systems such as face, fingerprint, and hand geometry recognition systems [14].

There are many factors which affect iris recognition performance. The resolution of iris images is important. International Standard Organization (ISO) / International Electrotechnical Commission (IEC) 19794-6 describes the iris image resolution necessary for recognition [16]. An iris diameter of more than 200 pixels is considered to be of “good” quality; 150 to 200 pixels is “acceptable”; and 100 to 150 pixels is “marginal” [15,16].

Because defocused iris images degrade the recognition rate, focused iris images are important to ensure good recognition performance. With portable iris recognition systems, optical and motion blurring are more severe than in conventional iris systems which are set up at an indoor fixed position.

Although ambient light exists, the iris patterns of Orientals are invisible because the amount of melanin is greater than in the irises of Westerners. Therefore, conventional iris recognition systems use NIR light illuminators so that iris patterns can be observed without causing any disturbance to user’s eye. In these systems, a NIR pass filter is attached to the front of the camera lens. Conventional iris systems for door access control, such as the LG IrisAccess 4000 [5] and the Panasonic BM-ET200 [7], have mainly been used indoors. Therefore, the captured iris images are only illuminated by the NIR illuminator of their system. However, a portable iris system can be used in outdoors where sunlight can affect the quality of the input iris images.

Capture volume is defined as the size of 3 dimensional space in which an eye must be placed for the iris system to reliably capture iris images of sufficient quality [15]. Capture volume relates to usability. In general, if the capture volume is large, the iris recognition system is easier to be used. The depth of the capture volume is determined by the depth of field (DOF) for fixed focus iris cameras. DOF can be defined as the Z-distance range of the subject while capturing a focused image.

Our portable iris system was designed considering the resolution of iris images, reduced optical and motion blurring, reduction of the sunlight effect, and an increase in capture volume. Further details are given in Sections 3.1 and 3.2.

3. THE PROPOSED PORTABLE IRIS SYSTEM

3.1. Design of portable iris recognition system

We used a Sony VGN-UX17LP UMPC as the processing unit for our system [12], which includes an Intel core 1.20 GHz processor, 512 MB SDRAM, and a 30 GB hard disk. The conventional size of one user’s iris template was 1,024 bytes [4,17-19], so more than 29,000,000 iris templates could be saved in our system. The PierTM 2.4 system only allows 200,000 iris templates to be saved [9], and the HIIDE system only allows 10,000 biometric templates, including two iris templates, ten fingerprints, and one facial image [10].

For the iris camera, we used a universal serial bus (USB)-type charge-coupled device (CCD) camera, which captures an iris image of 640×480 pixels at 15 frames/s. Fig. 2 shows the proposed system, which includes one iris camera with fixed focusing zoom lens and NIR light illuminators. By using a small zoom lens, our system meets the first requirement (resolution of the iris image). The average diameter of the iris images captured by the proposed system was 200 ~ 220 pixels within a Z distance operating range of 14 ~ 21 cm (details are explained in Section 3.2). According to ISO/IEC 19794-6 [16], our iris images met the size requirement for good quality qualified iris image resolution. Using the zoom lens, we can increase the Z-distance between the camera and eye, thereby improving the user’s convenience.

To meet the second requirement (reduced optical and motion blurring), we reduced the size of the lens hole of the iris camera, which increased the F-number and DOF. The maximum power of the NIR illuminator is limited by the requirement for eye safety, as shown in Section 4.3.

Matey *et al.* stated that “The capture volume is the three-dimensional spatial volume into which an eye must be placed and held for some period of time in order for the system to reliably capture an iris image of sufficient quality for iris recognition.” [15]. Based on this definition, the capture area (near the camera) of our system was measured as 3 (width) × 3 (height) cm² at a Z-distance of 14 cm. The capture area (far from the



Fig. 2. Proposed portable iris recognition system.

camera) was $4 \text{ (width)} \times 4 \text{ (height)} \text{ cm}^2$ at a Z-distance of 21 cm. The resulting depth of the two capture areas is 7 cm (21 – 14 cm). The user's eye can be positioned in the capture volume which is determined by the two capture areas.

The capture volume of our system is larger than portable iris systems such as the PierTM 2.4 system [9] and HIIDE [10]. In the case of PierTM 2.4, the capture area (near to the camera) was $3 \text{ (width)} \times 3 \text{ (height)} \text{ cm}^2$ at a Z-distance of 10.2 cm, the capture area (far from the camera) is $4 \text{ (width)} \times 4 \text{ (height)} \text{ cm}^2$ at a Z-distance of 15.2 cm, and the depth between two capture areas is 5 cm (15.2 – 10.2 cm). In the case of HIIDE, the capture area (near to the camera) was $3 \text{ (width)} \times 3 \text{ (height)} \text{ cm}^2$ at a Z-distance of 20.3 cm, the capture area (far from the camera) is $4 \text{ (width)} \times 4 \text{ (height)} \text{ cm}^2$ at a Z-distance of 25.4 cm, and the depth between two capture areas is 5.1 cm (25.4 – 20.3 cm). The width and height of two capture areas are solely determined by the zooming factor of camera lens. To increase the capture volume of our system, we increased the depth of two capture areas using an iris restoration algorithm. Details are discussed in Section 3.2.

3.2. Increasing DOF by iris restoration algorithm

The DOF can be calculated based on the F-number and camera focal length in conventional camera optics. In iris cameras, because the DOF means the Z distance operating range in which the captured iris image is identified, we measured the DOF in terms of recognition [15]. As previously mentioned, the longer the DOF is, the larger the capture volume becomes.

As shown in Fig. 3, we calculated the Hamming distance (HD) between the enrolled and recognized iris codes to measure the DOF. The proposed system uses a Gabor wavelet filter, and the extracted iris feature value was represented as a binary number by quantization (the algorithm is shown in Section 3.3.2). Then, we calculated the HD based on the eXclusive OR (XOR) operation between the enrolled and recognized codes. If the HD was smaller than a predetermined threshold (which was determined experimentally as 0.34), the input iris image was accepted as a genuine.

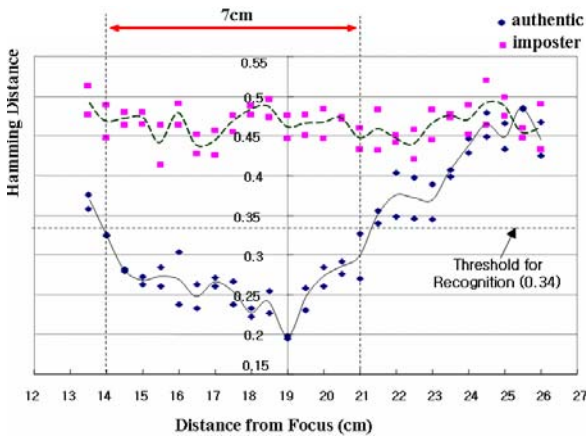


Fig. 3. The change of HD for measuring the DOF.

If the HD was greater than the threshold, the input image was rejected as an imposter (see Section 3.3.2). From this result, when the input image was captured in the Z distance range of 14 ~ 21 cm, the HDs of the genuine user's iris images were less than 0.34 and they were recognized. However, the HDs of an imposter's iris images were not affected by the Z-distance from focus as shown in Fig. 3. Here, genuine means a user whose iris is already enrolled and imposter means a user whose iris is not enrolled.

The DOF determines the depth of the capture volume, which relates to usability. To enhance usability, we significantly increased the DOF using the image restoration algorithm. To increase the DOF of iris cameras, Gracht *et al.* proposed an iris image restoration method which used a cubic phase modulation filter with which the modulation transfer function (MTF) could be changed according to the degree of blurring, and the focused iris image could be obtained by restoring the iris image with a Wiener filter [23,24]. However, because both the normalized power spectrum of the original image and the amount of noise for Wiener filtering had to be computed, the procedure required too much processing time to be used for real-time iris image restoration. In addition, the method used additional hardware called a cubic phase modulation filter and did not consider the changes of the point spread function (PSF) according to the degree of blurring [24].

To overcome such problems and to increase processing speed, we determined the parameters of the PSF based on the measured focus score and the camera calibration parameters [24]. Because the basic shape of the PSF was determined as a Gaussian function and only the parameters of the Gaussian function were calculated based on the measured focus score and the camera calibration parameters, we obtained the PSF at high speed. Based on the PSF and constraint least square (CLS) algorithm, we restored the input blurred image at real-time speed [24]. The resulting extended DOF became 12~23 cm. Fig. 4 shows the procedure of recapturing iris image, iris image restoration, and iris recognition. In a conventional system, the part of “ $T_1 < \text{The Focus Score} < T_2$ ” and “Restoring Iris Image” of Fig. 4 are not used. Thus, these two parts represent the technical advance of the proposed system. After capturing the iris image, our system measured the focus score of the captured image by convolution with a 5×5 pixel kernel, which is a high-pass filter as shown in Fig. 5 [24]. Because a focused image contains more high frequency components than a defocused one, the total power k of the high-frequency components can be used as an indication of the focus score.

$$FV = \frac{100k^2}{k^2 + c^2} \quad (1)$$

The total power k was normalized using (1), and we obtained a focus score having a range from 0 to 100. Here, 100 means the most focused image and 0 means the least focused image.

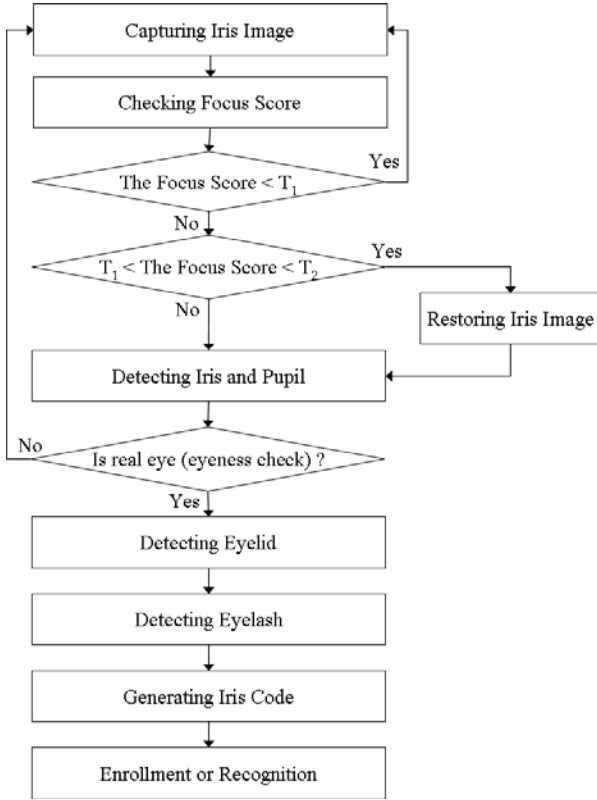


Fig. 4. The overall procedure of recapturing iris image, iris image restoration, and iris recognition.

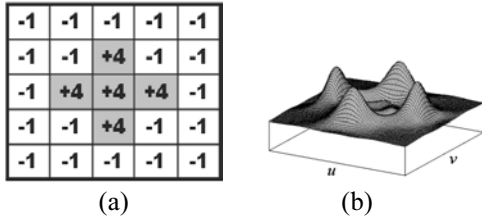


Fig. 5. The convolution kernel for focus score [24] (a) 5 × 5 pixel convolution kernel, and (b) power spectrum of (a).

Using the measured focus score, our system was able to determine whether or not the captured iris image could be used for iris recognition. As shown in Fig. 4, if the measured focus score was higher than T_2 , the proposed algorithm was able to determine that the captured image was focused and could be used for iris recognition. If the measured focus score was lower than T_1 , the proposed algorithm determined that the captured image was severely defocused and captured the input image again. If the measured focus score was between T_1 and T_2 , our system restored the captured iris image using the image restoration algorithm [24]. T_1 and T_2 were determined by experiment to be 50 and 70, respectively, which produced the highest recognition performance.

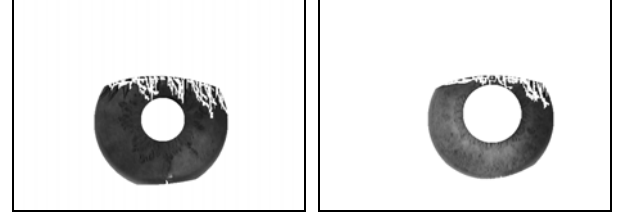
3.3. Iris recognition algorithm

3.3.1 Iris region segmentation

The captured iris images showed the entire human eye including the pupil, iris, sclera, eyelids, and eyelashes.



(a) Original images.



(b) Localized regions.

Fig. 6. Examples of the detected iris, eyelid, and eyelash regions.

To reduce detection errors of the iris region, the proposed algorithm used two circular edge detectors for locating the inner and outer boundaries of the iris at the same time [25]. We then obtained the center position and radius of both the pupil and the iris region. The proposed algorithm examined whether or not specular reflections existed in the detected pupil area. This process was named the “eyeness check.” In case of live eye, specular reflections by NIR illuminator appeared on the cornea surface of pupil region theoretically [26]. We checked whether corneal specular reflections existed in the pupil area of captured images. In the case of a fake eye such as a printed iris or closed live eye, there were no specular reflections in the pupil area. In this case, our system did not proceed for iris identification and recaptured the input image as shown in Fig. 4.

In the next step, the proposed algorithm detected eyelashes and eyelids. To detect eyelids, the proposed algorithm extracted the candidate points of the eyelid using eyelid detecting masks, and detected accurate eyelid regions using a parabolic Hough transform [27]. To detect eyelashes, the proposed algorithm used eyelash masks [28]. Fig. 6 shows examples of the detected iris, eyelid, and eyelash regions.

3.3.2 Extracting iris codes and recognition

Next, the proposed algorithm used a conventional Gabor filter-based method [1,17-19,29] and generated iris binary codes of 512 bytes (256 iris codes + 256 valid mask codes) per user. The valid mask codes represented whether or not the corresponding iris code was valid. For example, when an iris code was extracted from the occluded region of eyelid or eyelash, it was regarded as an invalid code. Only the valid iris codes were used for iris code matching with the enrolled codes.

The frequency and kernel size of the Gabor filter were determined to produce the minimum equal error rate (EER) with 1,010 iris images captured by our system. These were selected as 1/20 (frequency) and 25 pixels (kernel size), respectively. For the recognition test, we

excluded the 1,010 images used in determining the optimal frequency and kernel size, and instead used newly obtained images from 101 subjects (see Section 4).

To calculate the dissimilarity between the enrolled iris binary code vector (*codeA*) and the recognized vector (*codeB*), we used the HD based on the XOR operation [17-19]:

$$HD = \frac{\|(\text{code } A \otimes \text{code } B) \cap \text{validA} \cap \text{validB}\|}{\|\text{validA} \cap \text{validB}\|}, \quad (2)$$

where *validA* and *validB* denote vectors representing the valid codes of enrolled iris binary codes and recognized iris binary codes, respectively. In this case, only the valid iris code was used for calculating the HD [1,17-19]. If the calculated HD was lower than the predetermined threshold (determined to be 0.34 by experiment), the input iris was accepted as genuine. If it exceeded the threshold, the input iris was rejected as unenrolled.

4. EXPERIMENTAL RESULTS

We tested the proposed system with 101 subjects. All the test subjects were instructed to remove their glasses or sunglasses only for enrollment, but were permitted to wear them for identification. The subjects did not remove their contact lenses for enrollment or identification. Fig. 7 shows an example of iris images captured by our system. We performed the following objective and subjective tests.

4.1. Objective tests

In contrast with conventional iris systems for door access control, two kinds of users should be considered for a portable iris system. One is a “tester” (who holds the iris system), and the other is a “testee” (whose iris is to be identified).

For enrollment or identification, the tester held the iris camera to focus on the testee’s iris. During that process, the testee also tried to align his or her eye into the iris camera by using a cold mirror which was attached on the camera lens. (A cold mirror reflects visible light images

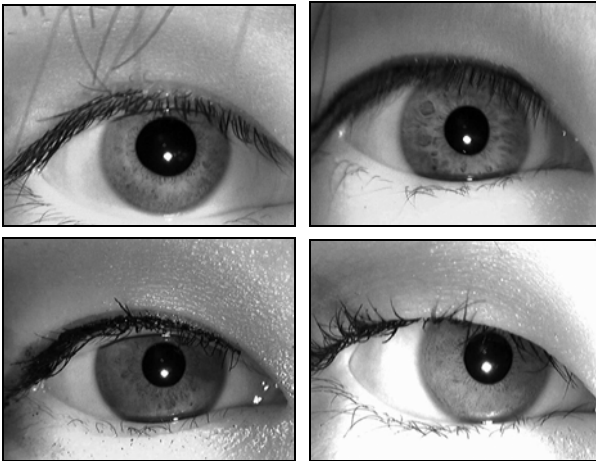


Fig. 7. Example of captured iris images.

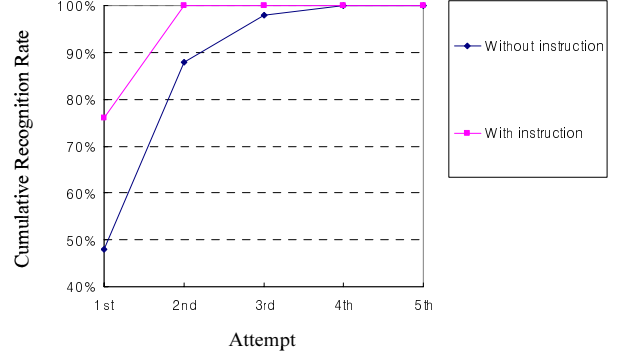


Fig. 8. Cumulative recognition rates.

and passes infrared light through the camera.) Thus, the testee could easily align his or her eye with the iris camera. Also, because the image was displayed on the UMPC screen, it was easy for the tester to align the testee’s eye. Our results showed that all the genuine users were successfully identified within five attempts, as shown in Fig. 8.

In previous studies, a cumulative rate was frequently used to test the operation of the biometric system [15]. It showed the correct recognition rate of a genuine user according to the number of identification attempts. For example, with instruction, our cumulative rate was 76% in the first trial and 100% in the second trial, as shown in Fig. 8.

This means that 76% of all the genuine users could be identified in the first trial with proper instruction, and the remaining 24% of the genuine users could be identified in the second trial. Thus, all of the genuine users could be identified within two identification trials. Because many attempts can cause much inconvenience to the user, the cumulative rate is frequently used for the operation test of a conventional biometric system.

At first, the testers used the system without instruction. The testers with no knowledge of iris recognition experienced difficulties since they did not know that they had to focus on only one iris of the testee. Some of them tried to focus on both irises, which caused misalignment errors. Conducting experiments without instructions caused the cumulative recognition rate to be as low as 48% at the first attempt, as shown in Fig. 8. However, the rate was greatly enhanced after three attempts, as shown in Fig. 8. In the next tests, we gave the testers operating instructions before they used the system. In this case, the cumulative recognition rate was 76% for the first attempt, and increased to 100% for the second attempt, as shown in Fig. 8. That means that individual identification was successfully completed during the first or second attempt.

Other factors decreased recognition rate. Some testees looked in the other direction (off-angle), opened their eyes too narrowly, or made facial movements.

To increase recognition performance, we told the testees to look at the camera lens, open their eyes widely, and keep their heads still. We also told the testers to focus on only one eye instead of both. As shown in Table 1, the recognition rates were much improved compared

Table 1. Improvement in success rates for the five least successful subjects.

Testee(1)-# Tester(2)-#	Change	Success Rate	
		Without instruction	With instruction
1 - 07	Eyes wide open	87%	100%
1 - 09	Looking at the camera lens	23%	100%
1 - 12	Still head	72%	100%
2 - 02	With instruction	12%	100%

to the rates without such instruction. In the case of “1-09” in Table 1, when the testee did not look at the iris camera lens, the recognition rate was 23%. However, when he looked at the camera according to instructions, the success rate increased to 100%. All of “1-07”, “1-09”, and “1-12” were the cases of the testee and “2-02” was the case of the tester. When “2-02” did not receive instruction in using our iris system, the success rate was 12%. However, with instruction on how to use our system, the rate became 100%.

Irrespective of whether users wore glasses, sunglasses, or contact lens, the recognition performances were unchanged. All the users were enrolled indoors. The recognition tests were performed in both indoors and outdoors. In the case of outdoors, we did not include an environment where much sunlight lit the user’s iris directly. Results showed that indoor recognition performance was almost the same as outdoor recognition performance.

In the next experiment, we measured the processing time of the proposed system which was, on average, 986.2 ms per person. Fig. 9 shows examples of using the proposed system. The system could be used in a verification mode with a user’s ID number, or in an identification mode without an additional ID number.

4.2. Subjective tests

We then performed subjective tests such as a usability evaluation. To evaluate usability, we designed a survey form including four categories based on Nielsen’s ten usability heuristics [30]. We performed subjective tests with the 101 subjects (including both testers and testees). The survey form measured the following four items:

- 1) User acceptance: how “easily” the user can use the system.
- 2) User convenience: how “conveniently” the user can use the system.
- 3) User satisfaction: the degree of “satisfaction” for



Fig. 9. Examples of using proposed system.

using the system.

- 4) User resistance: the degree of “resistance” to the infrared light illuminators of our iris system.

Users could select one of three scores (1, 2, or 3). In the case of user acceptance, convenience, and satisfaction, 3, 2, and 1 represented “very good,” “well,” and “not good (bad),” respectively. In case of user resistance, 3, 2, and 1 represented “no resistance,” “a little resistance,” and “much resistance,” respectively. Fig. 10 shows the results of the subjective tests.

Experimental results showed that users of 97% answered that user acceptance was “very good” or “well”. Users of 90% answered that user convenience was “very good” or “well”. Users of 91% answered that user satisfaction was “very good” or “well”. And users of 84% answered that user resistance was “much resistance” and “a little resistance.” Then, we performed the subjective tests with or without instructions. As shown in Figs. 11 and 12, user acceptance, convenience, satisfaction, and resistance were enhanced after receiving instructions.

To examine usability factors, the proposed system used a cold mirror in front of the iris camera lens. And the proposed iris system shows the focus score of every input image through user interface program.

The cold mirror helps a testee align his or her eye into the iris camera while a tester aligns the eye of the testee into the iris camera. With the cooperation of the testee and the tester, the proposed system can capture iris images with high acceptance and convenience. Based on

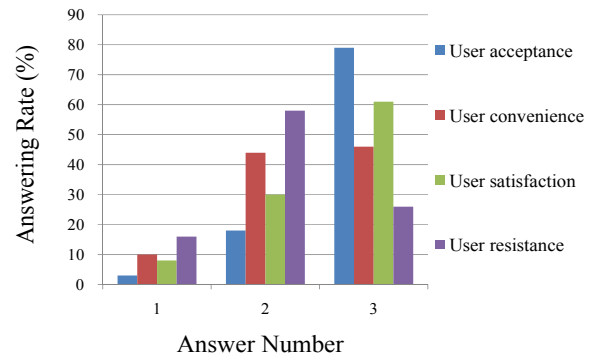


Fig. 10. Subjective tests including cases with and without instruction.

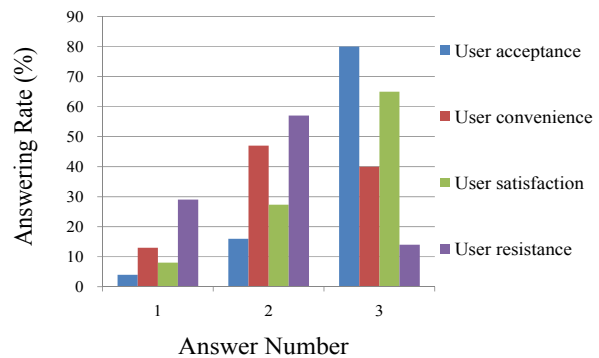


Fig. 11. Subjective tests without instruction.

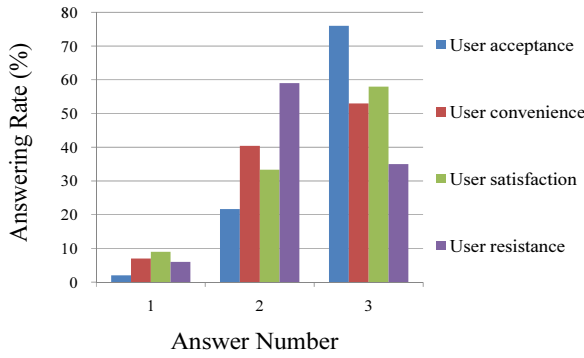


Fig. 12. Subjective tests with instruction.

the indicated focus score in the user interface program, the tester can easily obtain a focused eye image of the testee. The proposed system is also designed to meet the eye safety requirement as shown in Section 4.3, which can reduce the user resistance much.

The subjective tests in Section 4.2 show that the proposed system is designed to enhance user acceptance, convenience, and satisfaction, and to reduce resistance.

4.3. Eye safety test

We measured the eye safety of the NIR illuminator based on the American Council of Government and Industrial Hygienists (ACGIH) and threshold limit values (TLV) [15,20]. The ACGIH exposure limit for infrared radiation is defined by the following equation. For exposures greater than 1,000 s, irradiance must be limited to less than 10 mW/cm^2 [15,20]:

$$\sum_{770\text{nm}}^{3000\text{nm}} E_{\lambda} \cdot \Delta\lambda \leq 1.8t^{-3/4} \text{ W/cm}^2, \quad (3)$$

where λ represents the wavelength of incident light, summation is over the λ range where the light level is significant, E_{λ} represents the irradiance into the eye in W/cm^2 , and t represents the exposure time in s.

In the proposed system, the exposure time t was a maximum of five seconds (time-out) for enrollment or recognition. Thus, we obtained the maximum ACGIH exposure limits for infrared radiation as 540 mW/cm^2 based on Eq. (1). The experimental results showed that the infrared radiation power (0.53 mW/cm^2) of the proposed system was much less than the limit, so the proposed system met the safety requirements.

5. CONCLUSIONS

We proposed a portable iris recognition system that is more extensible than previous portable systems. In addition, the capacity of the proposed system for storing iris templates is much greater than previous systems. Based on tests, we found that the proposed system has a high level of usability.

In future work, we plan to combine iris and face recognition in our portable system to significantly increase recognition accuracy. Also, we plan to further increase usability.

REFERENCES

- [1] J. Daugman, "High confidence visual recognition of persons by a test of statistical independence," *IEEE Trans. on Pattern Analysis and Machine Intelligence*, vol. 15, pp. 1148-1161, 1993.
- [2] J. C. Yang and D. S. Park, "Fingerprint verification based on invariant moment features and nonlinear BPNN," *International Journal of Control, Automation, and Systems*, vol. 6, no. 6, pp. 800-808, 2008.
- [3] H. Kim, J. Choi, and M. Kim, "Human-robot interaction in real environments by audio-visual integration," *International Journal of Control, Automation, and Systems*, vol. 5, no. 1, pp. 61-69, 2007.
- [4] H. Lee, S. Hong, I. F. Nizami, and E. Kim, "A noise robust gait representation: motion energy image," *International Journal of Control, Automation, and Systems*, vol. 7, no. 4, pp. 638-643, 2009.
- [5] IrisAccess 3000 & 4000, <http://www.lgiris.com> (accessed on August 17, 2009).
- [6] IrisPass-M, <http://www.oki.com/jp/FSC/iris/en/> (accessed on August 17, 2009).
- [7] BM-ET 200, <http://www.panasonic.com/business/medicalvideo/biometrics.asp> (accessed on August 17, 2009).
- [8] IrisPass-H, http://www.pro-4-pro.com/media/downloads/product/9354428_1_p4p0804.pdf (accessed on August 17, 2009).
- [9] Pier™ 2.4, <http://www.l1id.com/pages/48-pier-2-4> (accessed on August 17, 2009).
- [10] HIIDE™ 4, <http://www.l1id.com/pages/47-hiide-series-4> (accessed on August 17, 2009).
- [11] "U.S. is building database on Iraqis - Biometrics key part of tracking suspects," *USA Today*, July 13, 2007.
- [12] VGN-UX17LP, <http://www.funponse.com/blog/computers/sony-vaio-vgn-ux17lp-mobile-pocket-pc.html> (accessed on August 17, 2009).
- [13] International Biometrics Group, "Independent testing of iris recognition technology (final report)," Rep. NBCHC030114/0002, May 2005, study commissioned by the U. S. Dept. Homeland Security.
- [14] T. Mansfield, G. Kelly, D. Chandler, and J. Kane, "Biometric product testing final report," http://www.cesg.gov.uk/policy_technologies/biometrics/media/biometricstestreportpt1.pdf (accessed on August 17, 2009).
- [15] J. R. Matey, O. Naroditsky, K. Hanna, R. Kolczynski, D. LoIacono, S. Mangru, M. Tinker, T. M. Zappia, and W. Y. Zhao, "Iris on the move: acquisition of images for iris recognition in less constrained environments," *Proc. of IEEE*, vol. 94, pp. 1936-1947, 2006.
- [16] ISO/IEC 19794-6, "Information technology. biometric data interchange formats - iris image data," 2005.
- [17] J. Daugman, "Demodulation by complex-valued wavelets for stochastic pattern recognition," *International Journal of Wavelets, Multi-resolution*

- and Information Processing, vol. 1, pp. 1-17, 2003.
- [18] J. Daugman, "The Importance of being random: statistical principles of iris recognition," *Pattern Recognition*, vol. 36, pp. 279-291, 2003.
- [19] J. Daugman, "How iris recognition works," *IEEE Trans. on Circuits and Systems for Video Technology*, vol. 14, pp. 21-30, 2004.
- [20] TLV of eye safety from ACGIH, <http://www.acgih.org> (accessed on August 17, 2009).
- [21] V. Ekaterina *et al.*, "Estimation of melanin content in iris of human eye," *Proc. of SPIE*, vol. 5688, pp. 302-311, 2005.
- [22] A. Vogel *et al.*, "Optical properties of human sclera, and their consequences for transscleral laser applications," *Lasers in Surgery and Medicine*, vol. 11, no. 4, pp. 331-340, 1991.
- [23] J. van der Gracht, V. P. Pauca, H. Setty, R. Narayanswamy, R. J. Plemmons, S. Prasad, and T. Torgersen, "Iris recognition with enhanced depth-of-field image acquisition," *Proc. of SPIE*, vol. 5438, pp. 120-129, 2004.
- [24] B. J. Kang and K. R. Park, "Real-time image restoration for iris recognition systems," *IEEE Trans on Systems, Man and Cybernetics - Part B*, vol. 37, no. 6, pp. 1-13, 2007.
- [25] D. H. Cho, K. R. Park, D. W. Rhee, Y. G. Kim, and J. H. Yang, "Pupil and iris localization for iris recognition in mobile phones," *Proc. of the SNPD*, Las Vegas, USA, 2006.
- [26] K. R. Park *et al.*, "A Study on iris localization and recognition on mobile phone," *EURASIP Journal on Advances in Signal Processing*, vol. 2008, pp. 1-12, 2007.
- [27] Y. K. Jang, B. J. Kang, and K. R. Park, "A study on eyelid localization considering image focus for iris recognition," *Pattern Recognition Letters*, vol. 29, issue 11, pp. 1698-1704, 2008.
- [28] B. J. Kang and K. R. Park, "A robust eyelash detection based on iris focus assessment," *Pattern Recognition Letters*, vol. 28, issue 13, pp. 1630-1639, 2007.
- [29] H. A. Park and K. R. Park, "Iris recognition based on score level fusion by using SVM," *Pattern Recognition Letters*, vol. 28, issue 15, pp. 2019-2028, 2007.
- [30] J. Nielsen, *Usability Engineering*, Morgan Kaufmann, 1994.



Young Kyoon Jang received the B.S. degree in Software from Sangmyung University in 2007. He has been the combined course of M.S. and Ph.D. from GIST since 2008. His research interests include biometrics systems, pattern recognition, and digital image processing.



Byung Jun Kang received the B.S. degree in Software in 2004 and the M.S. and Ph.D. degrees in Computer Science in 2006 and 2009, from Sangmyung University. He has been a senior researcher in Electronics and Telecommunications Research Institute (ETRI). His research interests include iris image restoration, iris recognition, biometrics, digital image processing, and pattern recognition.



Kang Ryoung Park received the B.S. and M.S. degrees in Electronic Engineering from Yonsei University in 1994 and 1996, respectively. He received the Ph.D. degree in Electrical and Computer Engineering from Yonsei University in 2000. He has been an associate professor in the Department of Electronics Engineering at Dongguk University since March 2008. And he is also a research member in the Biometrics Engineering Research Center (BERC). His research interests include face, iris, and finger vein image processing, and gaze detection.